

Due on Tuesday, February 13th.

If pictures are needed/relevant, please provide them with your solutions.

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1. [10 pts.] A charged sphere has radius a_0 and volume charge density

$$\rho = \begin{cases} \rho_0(r/a_0)^2 & \text{when } r \leq a_0 \\ 0 & \text{when } r > a_0 \end{cases}$$

where r is the distance from the center of the sphere. What is the total charge on the sphere?

Note: The volume charge density, also known just as the charge density, is the charge per unit volume.

Hint: it might help to consider the charge on a shell of infinitesimal width, and then integrate.

2. Electric Dipole.

In this problem, we ignore the z -direction. All points are on the xy plane.

A positive charge q is placed at position $(0, +d/2)$, and a negative charge $-q$ is placed at position $(0, -d/2)$. This is an *electric dipole*.

- (a) [2 pts.] For this electric dipole, what is the magnitude of the dipole moment? What is the direction of the dipole moment?
- (b) [5 pts.] Calculate the electric field at the point $(x_0, 0)$ on the x -axis, by calculating the fields due to each charge and adding them. This would be vector addition. Make sure to show pictures if relevant, indicating directions and angles.
- (c) [5 pts.] Calculate the electric potential $V(x, y)$ at an arbitrary point on the xy plane, by scalar addition of the contributions from the two charges.
- (d) [5 pts.] Use your expression for $V(x, y)$ to calculate the electric field at the point (x, y) , using the relationship between $\mathbf{E}$ and V .

Check that this yields the correct answer at the point $(x_0, 0)$, where you have already calculated the electric field by vector addition.

Note: the z -direction does not affect the answer, so you can neglect the z -derivative in the gradient.

3. Line (thin rod) of continuous charge.

In this problem also, we ignore the z -direction. All points are on the xy plane.

Consider a uniformly charged rod of length L , lying along the y -axis. The endpoints are at $(0, -L/2)$ and $(0, +L/2)$ and the center is at the origin. The linear charge density (charge per unit length) is λ .

- (a) **[1 pt.]** What is the total charge on the rod?
- (b) **[2 pts.]** Consider the point P on the x -axis, with coordinates $(x_0, 0)$. We want to calculate the electric field at this point. By symmetry, which direction should the field point towards?
- (c) **[7 pts.]** Take an infinitesimal element of the rod, between point $(0, y)$ and point $(0, y + dy)$. What is the charge within this element? What is the electric field contribution at point P due to this infinitesimal charge element? Evaluate separately the x - and y -components, dE_x and dE_y .
- (d) **[6 pts.]** Using the expressions for dE_x and dE_y , calculate the total electric field components at P by adding up contributions from all infinitesimal elements. This means integrating from $y = -L/2$ to $y = +L/2$.

It might help to know that

$$\int \frac{du}{(u^2 + a^2)^{3/2}} = \frac{u}{a^2(u^2 + a^2)^{1/2}}$$

- (e) **[3 pts.]** Simplify the expression for the case $x_0 \gg L$. Is the expression similar to the expression for a point charge? Explain why you could have expected this.
- (f) **[4 pts.]** Now imagine that the rod is infinitely long, i.e., $L \rightarrow \infty$. By taking the limit of the expression derived for finite L , calculate the electric field in this limit.

Comment: The electric field for an infinite line of charge (if you did this correctly) is inversely proportional to x_0 , not to the square of x_0 . The decrease of electric field with increasing distance is *slower* than compared to the electric field due to a single charge. Please think about whether this makes sense physically.